

Multi-Label Transfer Learning for Identifying Lung Diseases using Chest X-Rays

Azza El-Fiky
Department of Informatics
Electronics Research Institute
Cairo, Egypt
azzaelfiky@eri.sci.eg

Marwa Ahmed Shouman
Department of Computer Science and Engineering
Faculty of Electronic Engineering
Menoufia University
Minuf, Egypt
marwa.shouman@el-eng.menofia.edu.eg

Salwa Hamada
Department of Informatics
Electronics Research Institute
Cairo, Egypt
hesalwa@eri.sci.eg

Ayman El-Sayed
Department of Computer Science and Engineering
Faculty of Electronic Engineering
Menoufia University
Minuf, Egypt
ayman.elsayed@el-eng.menofia.edu.eg

Mohamed Esmail Karar
Dept. Industrial Electronics and Control Engineering
Faculty of Electronic Engineering
Menoufia University
Minuf, Egypt
mekarar@iecc.org

Abstract—Chest radiography presents one of the main medical imaging modalities for diagnosing lung diseases. To assist radiologists during interventional procedures, this paper aims at proposing a transfer learning-based classifier to automatically identify 14 different thoracic diseases in Chest X-ray (CXR) images. The proposed method is relied on deep residual neural networks with 50 layers (ResNet-50) to accomplish the diagnostic task of many chest diseases. In this study, a public dataset of 112,120 frontal radiograph images for Chest X-ray has been used for validating the proposed deep learning classifier. It achieved the best performance of multi-label classification of normal and 14 different lung diseases with an average area under curve (AUC) of 0.911 and F1-score of 0.66. This study demonstrated that the proposed ResNet-50 classifier as a transfer learning model outperforms other relevant methods in the previous studies for automatic multi-label classification of chest X-rays.

Keywords—X-ray, classification, Transfer learning, Thorax diseases, Computer-aided diagnosis.

I. INTRODUCTION

Lung diseases include several kinds of diseases like pneumothorax, Consolidation, lung cancer, infiltration, fibrosis, pneumonia, and emphysema [1]. Consolidation means that the lung density is increased focally [2]. Effusion is fluid in the gap around the lung, pneumothorax is when there is air in the pleural space around the lung [3]. Pneumonia is an infectious disease in one or both lungs that can be caused by viruses and bacteria [4] and caused 2.56 million deaths in 2017 [5] and lung cancer caused 2.09 million deaths worldwide in 2018 [6].

Chest radiography or Chest X-Ray (CXR) is one of the most popular types of medical imaging modalities for the diagnosis of many respiratory or thorax abnormalities as mentioned above [1]. With an increasing number of patients,

radiologists must overwork, and it may be difficult to diagnoses and report chest images timely. CXR and computed tomography (CT) images are the most methods used to diagnose lung diseases, CXR images are cheaper, faster, and easier than CT, and they are offered in smaller clinics. By using CXR, the patients have exposed too little radiation than CT, it is still the most needed radiological examination [8]. Therefore, it is important to help radiologists to diagnose and report X-ray in a short time, perform medical image classification accurately, and improve patient healthcare by using the computer-aided-diagnosis system (CAD) [9], [10]. CAD systems can be enhanced using Computer Vision (CV) and Deep Learning (DL) techniques.

Recently, deep learning and machine learning approaches have been utilized to classify medical images into normal and abnormal classes and help radiologists to focus on abnormal images, also can categorize medical images into more classes according to different diseases of the image, it is known as multi-label image classification [11]. The image classification problem performs in two steps: the first step is to extract features from the image through feature descriptors, the second step feed extracted features to a classifier like support vector machines (SVMs) algorithm. This is a traditional technique and called the machine learning technique and its accuracy depends on the feature extraction method used [12]. Another technique that has been developed for the image classification task is deep learning, it enables automatic feature extraction and classification in one step by modeling data via several processing layers [10],[12]. DL has been used in different fields such as Pattern Recognition, Machine Translation, Natural Language Processing and Image Classification [13]. DL models introduce a chance to expand research and apply the basis for classifying the thorax diseases using X-ray images. Most of the algorithms belong

to DL such as pre-trained Convolutional Neural Network models are required to accurately diagnose various types of diseases at the time.

The Convolutional Neural Network (ConvNet /CNN) was proposed in 1988 by Yann LeCun. It is widely used for image and video recognition, language translate, natural language processing also applied in several medical applications like lymph nodes detection, breast cancer diagnosing, skin cancer classification [14] and implant tracking in X-ray images during minimally invasive surgery [9]. The first CNN model (LeNet) was developed by LeCun et al. in 1998. LeNet model was used to read zip codes, digits, and handwritten character recognition problems [13]. The CNN layers are including convolution layers, pooling layers and fully connected layers.

One of the most obstacles of CNN models is that need a huge amount of data to achieve high performance, sometimes it may be difficult to obtain such as medical images. So, transfer learning is the first choice in medical image processing including, chest radiography. In transfer learning, it does not build the model from scratch but uses pre-trained CNN models such as that trained on a huge dataset such as ImageNet that contains more than one million images for 1000 class, to train new datasets by using the weights and network layers. The main advantages of transfer learning are used for training small datasets and reduced the training period [10], [14]. There are different strategies of transfer learning such as freezing CNN layers, fine-tuning and fixed feature extractor. The main contributions of this work are summarized as follow:

- Proposing a modified model of transfer learning as a radiological tool to assist clinicians for automated diagnosis of 14 thorax diseases, e.g., infiltration and effusion, in X-ray images.
- Using the transfer learning technique in conjunction with the ResNet-50 model pre-trained on ImageNet to enhance the performance of multi-label classification of thorax diseases in chest X-ray 14 dataset.
- Presenting a comprehensive experiment on the ChestXray14 dataset. The results demonstrated that the performance of our proposed method is better than other methods in the literature.

The remainder of this paper is structured as follows. Section II introduces an overview of related work. The materials methods are presented in Section III. The experimental results and comparison with previous studies are presented in Section IV. The discussion is presented in Section V. Finally, Section VI presents the conclusion and future work.

II. RELATED WORK

In recent years, many studies had conducted for the automatic classification and detection of thorax diseases using deep learning algorithms. Some of the previous studies presented as follows: Bharati et al. [15] suggested a new hybrid model by integrating VGG16, data augmentation and spatial

transformer network. The hybrid framework had used for detecting thoracic pathology in chest x-ray images. Yan et al. [16] suggested an annotation model that focuses on abnormal parts in CXR images by using a manual report annotated by a professional radiologist, employed LSTM to annotate the unlabeled CXR images in the dataset and classified abnormal X-ray images using DenseNet. CXNet-m1 is a hierarchical CNN architecture that used a sin-loss function to classify chest radiography into normal or abnormal classes [11].

Transfer learning is an effective technique for training small data, especially in the medical domain. The transfer learning technique with CNN pre-trained models such as DenseNet and ResNet was used in several studies for binary classification like pneumonia and consolidation and multi-label classification of chest diseases. Some of the previous studies presented as follows: In the study of [7], Varshni et al. appraised DenseNet-169 pre-trained model to extract features for analyzing CXRs, and then classified the extracted features with the different classifiers, like support vector machine, into pneumonia or normal. Khamparia et al. [17] proposed a hybrid transfer learning model using Modified VGG (MVGG), residual network (ResNet), mobile network (MobieNet) to detect breast cancer. In [18], using the transfer learning approach, Chouhan et al. suggested a model consist of five different convolutional neural networks pre-trained on ImageNet for feature extraction then combined the prediction of these five CNN by using ensemble classification and taking the majority voting as the final prediction for pneumonia detection. Allaouzi et al. [12], have used chest radiography images to detect chest diseases using pre-trained CNN as feature extractor and multi-label problem transformation (Binary Relevance (BR), Classifier Chain (CC) and Label Powerset (LP) with logistic regression (LR)) as the classifier of features extractor. Our proposed model used the transfer learning technique in conjunction with ResNet-50 model pre-trained on ImageNet to enhance the performance of multi-label classification of thorax diseases in the chest X-ray images.

III. MATERIALS AND METHODS

A. Dataset

Chest X-ray 14 dataset was proposed in 2017 by Wang et al. [1] that comprised 112,120 frontal radiograph images with disease labels from 30,805 unique patients. CXR dataset contains 14 chest diseases was including Pneumonia, Infiltration, Fibrosis, Effusion, Atelectasis, Pneumothorax, Nodule, Cardiomegaly, Mass, Consolidation, Emphysema, Pleural Thickening, Edema and Hernia. In addition to normality images, called 'No Finding' that hasn't labeled to these diseases. All images in the dataset have a size of 1024 x 1024 with an 8-bit greyscale value in a PNG format. Most of the images are labeled with more than one thorax disease

at the same time that makes it a typical problem of multi-label classification.

B. Transfer learning method

Medical image dataset usually is small and insufficient annotated to be utilized for training Convolutional Neural Networks (CNNs), so the technique of transfer learning has been used to avoid the problem of data shortage, also to decrease the time and the effort needed to construct a model from scratch for a particular problem. Transfer learning has used the knowledge of a model that trained on a huge dataset to train a small dataset [10].

The multi-label classification method is received increased attention and has applied in many real-world applications such as medical diagnosis, text categorization, sentiment classification and object tracking [21]. In the multi-label classification method, each instance can be associated with more than one class at the same time. There are two general classes of algorithms for multi-label classification, problem transformation which meant converting the task of multi-label classification into one or more binary classification tasks, and algorithm adaptation that modifying the single-label classifier and making it appropriate to deal with multi-label data directly [22]. Another category is an ensemble method that collects the outputs of several classifiers depend on either algorithm adaptation or problem transformation. In this work, we used transfer learning in conjunction with ResNet-50 model to enhance the performance of multi-label chest X-ray classification.

There is a different strategy of transfer learning, these strategies include freezing CNN network layers, fine-tune CNN network layers and fixed feature extraction. In freezing CNN network layers, the CNN weights of the first layers have been frozen and then train only some top layers that have specific features of the classes in the main dataset [12]. One type of pre-trained CNN model is Residual Network (ResNet). ResNet was developed by Kaiming He et al. in 2015 and achieved an error rate of 3.57% which is less than the previous network, it was used for classification and recognition tasks [23]. ResNet was built with several different numbers of layers, such as 18, 34, 50 and 152. The architecture of ResNet-50 was designed to avoid the problem that confronts the networks with an increasing number of layers by using skip connections. ResNet uses a skip layer connection that combines the Residual Network with multi-branch architecture and achieves better performance and faster convergence training [13].

The architecture of ResNet-50 is illustrated in Figure 1. ResNet-50 pre-trained model consists of 50 layers, these layers are 49 convolution layers (1 + 16x3) and fully connected layer, it has five blocks of convolution layers which called with Block1- Block5, followed by FC classifier which consists of global average pooling that provides better classification accuracy with faster computation and it used to decrease the number of parameters, then fully connected

layer with 15 dimensions for classifying 14 thorax diseases and normality label.

C. Proposed deep learning model

We proposed a transfer learning model with ResNet-50 for multi-label classification of chest radiography images. The hyperparameters of the proposed model are illustrated in Table 1. Figure 2 clarifies the block diagram of our proposed model that contains three steps: preprocessing CXR images, transfer learning with ResNet-50 pre-trained model and classification.

- Step #1: Preprocessing**

CXR images have been preprocessed in two steps. First, we resized the input images from 1024 x 1024 to 224 x 224 pixels, then we applied the data augmentation to increase the quality and the size of the data and avoid overfitting by applying random transformation on the available data samples. The images were augmented with random horizontal flipping and standard normalization.

- Step #2: Transfer learning model**

In this work, transfer learning with ResNet-50 model has been used for the multi-label classification of chest X-ray 14 dataset by fine-tuning all layers and substitute the fully connected layer with other fully connected layer using 15 neurons for classifying 14 pathology labels and normality label with a sigmoid activation function.

- Step #3: Classification**

Our proposed model is a multi-label classification that classifies the images into normality, one or more disease of 14 thorax diseases.

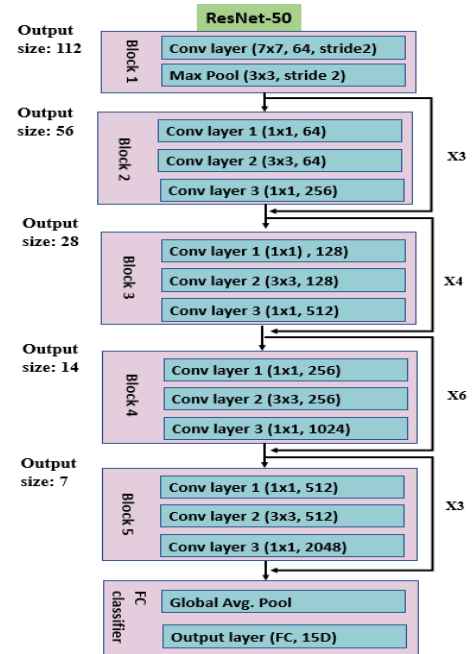


Figure 1. The basic architecture of ResNet-50 model.



Figure 2. The block diagram of proposed model for multi-label classification of CXR images.

Table 1. Hyperparameter values used for training the proposed model.

Parameters	Value
Batch size	32
Number of epochs	60
Optimizer	Adam
Learning rate	0.001
loss function	binary cross-entropy
Activation function	Sigmoid

D. Evaluation Metrics

Prediction of multi-label classification is usually classified into four groups: true positive (TP) is the number of samples that are positive and correctly classified to be positive, true negative (TN) is the number of samples that are negative and correctly classified to be negative, false positive (FP) is the number of samples that are negative and incorrectly classified to be positive and false negative (FN) is the number of samples that are positive and incorrectly classified to be negative [4].

Evaluation metrics of multi-label classifier can be categorized as example-based and label-based. Label-based metric evaluates the predictive performance for each class, then computes the performance average over all classes [24]. The average label-based metric can be macro-averaged or micro-averaged. The AUC (area under the receiver operating characteristic) is commonly used in multi-label classification, which shows the performance of classification at all classification thresholds, and it is depended on the relation between true positive rate and false-positive rate. The example-based evaluates the average differences between the actual and the predicted classes over all samples in the testing dataset [24]. The most common metrics used in the example-based are F1 score and hamming loss.

Precision (Positive Predictive Value) measures the ratio number of instances that correctly classified as positive to the total instances predicted as positive, precision is defined as

$$precision = \frac{TP}{(TP + FP)} \quad (1)$$

Recall (called True Positive Rate) measures the ratio number of instances which correctly classified as positive to the total instances that actual positive, recall is defined by (2)

$$recall = \frac{TP}{(TP + FN)} \quad (2)$$

F1-score is the harmonic mean of precision and recall, it transfers the balance between the precision and recall of the model, as represented in (3). The higher value of the F1-score indicates the higher performance of the learning model.

$$F1 - score = \frac{(2 * precision * recall)}{(precision + recall)} \quad (3)$$

Hamming loss is an evaluation metric which widely used in Multi-label classification. It represents the fraction of the incorrect labels to the total number of labels; hamming loss is given in (4).

$$Hamming\ loss = \frac{1}{NL} \sum_{i=1}^N \sum_{j=1}^L Xor(y_{i,j}, \hat{y}_{i,j}) \quad (4)$$

where N and L denotes the number of samples and number of labels, $y_{i,j}$ and $\hat{y}_{i,j}$ denotes the actual label and predicted label for the image.

IV. RESULTS AND EVALUATION

A. Experimental Setup

The main disadvantage of ChestX-ray14 is the imbalanced data in the number of individual classes that may lead to mislearning and classification error. A random 50000 sample of the dataset has been used to reduce the drawback of imbalance instance. Figures 3 show the frequency of diseases for each pathology class. The dataset was divided into 80% training set and 20% testing set. Before passing the images to the network, the images were resized into 224x224 and augmented. The experiments of this study were executed using the python programming language and Keras framework on the GPU Kaggle environment.

B. Evaluation Results

The main goal of using the transfer learning approach was automatically classifying multi-label chest radiography images. The assessment of x-ray images disease classification was done using AUC, Micro F1- score and hamming loss. Transfer learning and ResNet-50 achieved the AUC value of 0.911, Micro-F1 score value of 0.66 and hamming loss value of 0.404 which is still not better. The ROC curves and the AUC values for each classifier on the 14 pathology labels in addition to the normality label are illustrated in Figure 4. The AUC values are between 0.845 (for Infiltration case) and 0.943 (for Edema case) in all the ChestX-ray14 pathology labels.

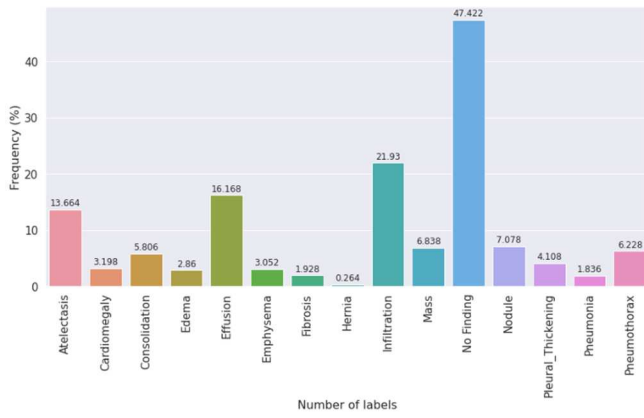


Figure 3. Frequency of diseases in each pathology class for 50000 random sample from Chest X-Ray 14.

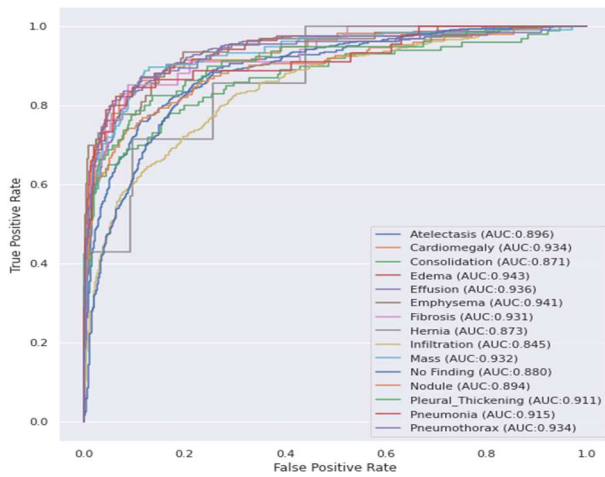


Figure 4. The ROC curves and AUC values on ChestX-ray14 dataset.

Table 2. Comparison the metrics of Avg. AUC, Micro-F1 and Hamming Loss on chest x-ray14 dataset.

Classifier	Metrics		
	Avg. AUC	Micro-F1	Hamming Loss
CheXNet [19]	0.8414	0.435	---
AG-CNN [20]	0.871	---	---
BR(LR) [12]	0.882	0.547	0.061
LP(LR) [12]	0.880	0.540	0.067
CC(LR) [12]	0.881	0.561	0.069
Our proposed	0.911	0.66	0.404

C. Comparison with Previous Studies

We compared the obtained results of our proposed ResNet-50 method with other results based on the chest x-ray14 dataset. The comparative results are illustrated in Table 2 and Table 3. In Table 2, the results of Avg. AUC of 14 pathology labels, Micro-F1 and Hamming Loss compared with the results of Rajpurkar et al. [19], Guan et al. [20] and Allaouzi et al. [12]. The proposed achieved result in Micro-

F1 and Avg. AUC increased by 1% (from 0.882 to 0.911) and 2.9% (from 0.561 to 0.66) respectively compared with the results of [12], but the Hamming Loss metric is not satisfactory.

Table 3 shows the Avg. AUC values for each thorax disease in the chest x-ray14 dataset, in comparison to [12] there are 11 out of 14 pathology labels was obtained the best AUC value, our proposed model achieved more 4% higher AUC value in Atelectasis, Infiltration, Nodule, Pneumonia, Fibrosis and Pleural Thickening and AUC value of the normal cases increased by 7.9% (from 0.811 to 0.88). Comparing our model with [20] there are 12 out of 14 diseases was achieved the best AUC value and Avg.AUC was increased by 4% (from 0.871 to 0.911). While comparing with [19] our result was obtained better classification AUC values in 13 out of 14 pathology labels and Avg.AUC was increased by 6.95% (from 0.8414 to 0.911).

V. DISCUSSION

The Deeper CNN networks like ResNet-50 model are able to achieve a good classification of lung disease classes, as depicted in Fig. 2. In Table 3, the transfer learning with ResNet-50 model can learn the features in X-ray images by fine-tuning the network layers. The advantage of ResNet-50 model is using a shortcut connection that is used to increase the performance of training and alleviate the problems that occur when the model has several layers such as degradation and vanishing gradient.

The main limitation of medical images such as CXR is the scarcity of available annotated data. Therefore, the transfer learning method has been used to overcome the problem of a limited amount of data. Also, using data augmentation to increase the number of annotated and improve the generalization of the model data by making several copies of the images using different transformations such as normalization and flipping. Another limitation of CXR in the chest X-ray 14 dataset is the imbalanced instance in the number of each label. The imbalanced data can be avoided by under-sampling most labels and oversampling the distribution minority of labels. However, the proposed deep learning classifier (ResNet-50) is still valid to achieve the best results than other methods in previous studies.

VI. CONCLUSIONS AND FUTURE WORK

Chest radiography is the most common way used to detect and classify several thorax diseases in the world. This study introduced a transfer learning model that was applied for automatic multi-label classification of 14 thorax diseases and normal case in chest X-ray 14 dataset. The presented method uses a transfer learning to use ResNet-50 CNN model pre-trained on the ImageNet dataset to improve the performance of chest diseases diagnose. The data augmentation was used to increase the performance of the model by using standard normalization and horizontal flip. Compared with previous works, the proposed model

Table 3. Comparison results of AUC values on all thorax disease in chest x-ray14 dataset.

Thorax disease	CheXNet[19]	AG-CNN[20]	BR(LR)[12]	LP(LR) [12]	CC(LR)[12]	Our proposed
Atelectasis	0.8094	0.853	0.855	0.854	0.855	0.896
Cardiomegaly	0.9248	0.939	0.952	0.948	0.952	0.934
Consolidation	0.7901	0.842	0.849	0.849	0.848	0.871
Edema	0.8878	0.924	0.931	0.931	0.931	0.943
Effusion	0.8638	0.903	0.911	0.910	0.909	0.936
Emphysema	0.9371	0.932	0.942	0.941	0.942	0.941
Fibrosis	0.8047	0.864	0.867	0.861	0.865	0.931
Hernia	0.9164	0.921	0.986	0.977	0.986	0.873
Infiltration	0.7345	0.754	0.749	0.748	0.747	0.845
Mass	0.8676	0.902	0.913	0.912	0.913	0.932
Nodule	0.7802	0.828	0.810	0.809	0.810	0.894
Pleural Thickening	0.8062	0.837	0.848	0.843	0.843	0.911
Pneumonia	0.7680	0.774	0.809	0.804	0.811	0.915
Pneumothorax	0.8887	0.921	0.929	0.928	0.929	0.934
Average AUC	0.8414	0.871	0.882	0.880	0.881	0.911
Normal	-----	-----	0.804	0.811	0.794	0.880

achieved the best performance score of Avg. AUC and Micro-F1 which is 2.9% and 1% higher than other comparable models. Furthermore, the proposed model can be extended to include other pneumonia diseases such as the COVID-19, as presented in Ref. [25].

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